

Network Topologies Lab - Cisco Packet Tracer

Learning Objectives

By the end of this lab, you will be able to:

1. Build and identify a star topology using a central switch and multiple end devices.
2. Build and identify a partial or full mesh topology and explain how physical redundancy creates alternate paths.
3. Compare client-server and peer-to-peer network models using the visual arrangement of devices in Packet Tracer.

Lab Safety and Best Practices

- Save your work frequently.
- Use clear device names and labels.
- Arrange devices neatly so the topology is easy to recognize.
- Use a network diagram or documentation template to record the physical design.
- Do not use real passwords or sensitive information.
- Make one physical change at a time and observe how it changes the diagram.

Activity 1: Building a Star Topology Network

Objective

Create and identify a basic star topology using one central switch and multiple end devices.

Network Design

Use:

- One switch: S1
- Four to six PCs: PC1 through PC4, or PC6
- Copper straight-through cables

Construction Steps

1. Open Cisco Packet Tracer.
2. Place one 2960 switch in the workspace.
3. Rename the switch S1.
4. Place four to six PCs around the switch.
5. Rename the PCs PC1 through PC4, or PC6.
6. Use copper straight-through cables to connect:
 - PC1 to an available port on S1
 - PC2 to an available port on S1
 - Continue for each PC
7. Arrange the devices so the switch is visibly in the center.
8. Make sure each cable is connected to a valid port on both devices.
9. Add text labels showing the device names.
10. Save the Packet Tracer file.

Observation Questions

Answer the following questions using the completed diagram:

1. Which device is the central device?
2. How many end devices are connected to the central device?
3. How many physical connections are shown?
4. What makes this design a star topology?
5. What would happen to the connected devices if the central switch failed?
6. What would happen if only one PC cable were disconnected?

Activity 2: Building a Mesh Topology Network

Objective

Create and identify a partial mesh topology while observing physical redundancy and alternate paths.

Recommended Design

Create a partial mesh using:

- Three switches: S1, S2, and S3
- At least one PC connected to each switch
- Three inter-switch links:
 - S1 to S2
 - S2 to S3
 - S1 to S3

This design creates a triangular mesh between the switches. The triangle shows more than one possible physical path between parts of the network.

Construction Steps

1. Place three switches in a triangular arrangement.
2. Rename them S1, S2, and S3.
3. Place at least one PC near each switch.
4. Rename the PCs clearly.
5. Connect each PC to its local switch.
6. Connect the switches using available Ethernet ports:
 - Connect S1 to S2.
 - Connect S2 to S3.
 - Connect S1 to S3.
7. Use clear labels for all links.
8. Arrange the devices so the triangular design is easy to see.
9. Make sure all cables are connected to valid ports.
10. Save the Packet Tracer file.

Observation Questions

Answer the following questions using the completed diagram:

1. How many switches are in the mesh design?
2. How many links connect the switches to one another?
3. How many possible physical paths appear to connect S1 and S2?
4. Which alternate path could connect S1 and S2 if their direct link were unavailable?
5. How does the mesh design provide more physical redundancy than a simple star?
6. What would happen to the overall design if one inter-switch cable were disconnected?
7. Why does a mesh require more cables and ports than a star?

Important: A physical connection does not always mean that every link forwards traffic at the same time. For this activity, focus on identifying the visible physical paths and redundancy shown in the diagram.

Activity 3: Client-Server vs Peer-to-Peer Comparison

Objective

Build two small networks and compare their visual device arrangements.

Part A: Client-Server Network

Use:

- One switch: S4
- Three PCs: PC7, PC8, and PC9
- One server: Server0

Construction Steps

1. Place one switch, three PCs, and one server.
2. Rename the switch S4.
3. Rename the PCs PC7, PC8, and PC9.
4. Connect each PC and the server to S4.
5. Arrange the server so it is visually distinct from the client PCs.
6. Add labels identifying the server and client devices.
7. Save the Packet Tracer file.

Observation Questions

1. Which device is visually different from the other devices?
2. Which devices appear to be clients?
3. How does the switch connect the server to the client devices?
4. What feature of the arrangement identifies this as a client-server model?
5. Why might an organization place a dedicated server in a central location?

Part B: Peer-to-Peer Network

Use:

- One switch: S5
- Three PCs: PC10, PC11, and PC12

Construction Steps

1. Place one switch and three PCs.
2. Rename the switch **S5**.
3. Rename the PCs **PC10**, **PC11**, and **PC12**.
4. Connect each PC to **S5**.
5. Arrange the PCs so they appear as similar devices connected through the switch.
6. Add labels identifying the peer devices.
7. Save the Packet Tracer file.

Observation Questions

1. How are the devices in this design similar?
2. Is there a separate dedicated server shown?
3. What visual feature identifies this as a peer-to-peer model?
4. How does this arrangement differ from the client-server arrangement?
5. Why might a small group choose a peer-to-peer design?